

FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Predictive Maintenance and Anomaly Detection Algorithm for Biomethane Plants

Mihai Adrian Cican



Mestrado em Engenharia e Ciência de Dados

Supervisor: Gil Manuel Gonçalves

Second Supervisor: Eliseu Pereira

Second Supervisor: João Reis

July 21, 2026

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Declaration of Honor

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Abstract

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Acknowledgements

Write your acknowledgements here. Do not forget to mention projects and/or grants that you have benefited from, institutions or companies that may have provided for conditions to enable or make your work more amenable, if any. Ask your supervisor about the specific textual format to use, as funding agencies are often strict about this.

Author

*“Until I began to learn to draw,
I was never much interested in looking at art.”*

Richard P. Feynman

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Abbreviations

AI	Artificial Intelligence
DL	Deep Learning
ML	Machine Learning
ANN	Artificial Neural Network
RNN	Recurrent Neural Network
MLP	Multilayer Perceptron
EDA	Exploratory Data Analysis
CV	Computer Vision
NLP	Natural Language Processing
PdM	Predictive Maintenance
RUL	Remaining Useful Life
AD	Anomaly Detection
FD	Fault Detection
CPS	Cyber-Physical Systems
GPU	Graphics Processing Unit
TPU	Tensor Processing Unit
IoT	Internet of Things
PSA	Pressure Swing Adsorption
VPSA	Vaccum Pressure Swing Adsorption
TSA	Temperature Swing Adsorption
ESA	Electric Swing Adsorption

Chapter 1

Introduction

1.1 Context

The 21st century has witnessed remarkable advances in Artificial Intelligence (AI) systems, reshaping numerous industrial and societal sectors. (Golec et al. (2025)). The introduction of AlexNet in 2012 (Krizhevsky et al. (2012)), AlphaGo's victory over Lee Sedol in 2016 (Silver et al. (2016)), and the public release of ChatGPT in 2022 (OpenAI et al. (2024)) are among the most significant milestones in modern AI, advancing the fields of Computer Vision (CV), Sequential Decision-Making, and Natural Language Processing (NLP), respectively. These breakthroughs were driven by a rapid growth in computational power, the development of specialized hardware such as GPUs and TPUs, and the widespread adoption of Deep Learning (DL) techniques (Sadjieh et al. (2026)).

As a result, DL has become the dominant paradigm in modern Machine Learning (ML) due to its ability to learn highly complex and abstract representations directly from data (LeCun et al. (2015)).

1.2 AI in Industry 4.0

Over the last fifteen years, the industrial sector has undergone a profound digital transformation, leading to the emergence of the Industry 4.0 paradigm. Industry 4.0 is characterized by the integration of advanced manufacturing technologies, Cyber-Physical Systems (CPS), the Internet of Things (IoT), cloud computing, and interconnected production environments. These technologies enable the continuous acquisition of large volumes of sensor data, creating smart factories capable of monitoring, communicating, and adapting their operation in real time.

The availability of these large-scale industrial datasets has created favourable conditions for the widespread adoption of DL techniques in this field. AI systems are increasingly being employed to support production planning and scheduling, process monitoring and control, Predictive

Maintenance (PdM), robotics, quality inspection, and process optimization, contributing to improved efficiency, reliability, and productivity in modern manufacturing environments (Cerqueus et al. (2025)).

1.3 Motivation

Despite the significant advances in AI technologies, the application of DL techniques in Anomaly Detection (AD), Fault Detection (FD), and PdM tasks for biomethane plants remains limited. This limitation highlights a gap in the literature regarding the development of advanced data-driven solutions for the biomethane upgrading industry. Furthermore, there is a lack of research on multi-task learning models capable of performing different types of predictions simultaneously in this industry. Therefore, the development and evaluation of such models represent a promising research direction. This thesis aims to train and evaluate different DL models (both multi-task and single-task) in order to investigate whether deep multi-task learning approaches can improve prediction performance regarding AD, FD, and PdM when compared with traditional ML models.

1.4 Objectives

The objectives of this thesis are:

- To perform an Exploratory Data Analysis (EDA) of the biomethane upgrading dataset in order to better understand the data, characterize the process variables and identify potential challenges for model development.
- To develop and train single-task DL models for AD, FD, and PdM.
- To develop and train multi-task DL models using multiple architectural configurations capable of jointly performing several prediction tasks.
- To develop and train traditional ML models for each prediction task, providing a baseline for comparison.
- To compare and evaluate the performance of all proposed models using appropriate evaluation metrics.
- To evaluate the robustness of the developed models.
- To investigate whether multi-task learning improves predictive performance compared with single-task DL models and traditional ML approaches.

1.5 Company Presentation

This dissertation was developed in collaboration with Sysadvance, a Portuguese company specialized in gas separation and purification technologies. The company develops industrial solutions

for gas generation, purification, and biogas upgrading, providing the real industrial case study considered throughout this work.

1.6 Thesis Structure

This dissertation is organized into XX chapters.

- **Chapter 1** introduces the research context, motivation, objectives, and overall organization of the dissertation.
- **Chapter 2** presents the theoretical background required to understand the problem addressed in this work. It introduces the concepts of biogas and biomethane production, biogas upgrading technologies, the considered Vacuum Pressure Swing Adsorption (VPSA) plant, the operating principle of the vacuum pump, and the monitored instrumentation.

Chapter 2

Background

This chapter introduces the technological background required to understand the industrial process addressed in this dissertation. It first presents the fundamentals of biogas production and upgrading technologies, followed by a description of the VPSA process employed in the considered biomethane plant. Finally, the vacuum pump, which constitutes the main asset analysed in this work, is presented together with its operating principle, instrumentation, maintenance procedures, and common faults.

2.1 What is Biogas?

Biogas is a renewable gaseous fuel produced through the anaerobic digestion of biodegradable organic matter, a biological process in which microorganisms decompose organic compounds in the absence of oxygen. This process converts complex organic materials into simpler compounds while producing a gaseous mixture known as biogas. A wide range of feedstocks can be used for anaerobic digestion, including agricultural residues, livestock manure, municipal solid waste, sewage sludge, food waste, and other types of organic biomass, as illustrated in Figure 2.2.

The composition of biogas depends on the type of feedstock and the operating conditions of the anaerobic digestion process. It is primarily composed of methane (CH_4), typically accounting for 50–80% of its volume, and carbon dioxide (CO_2), which represents approximately 20–50%. In addition to these major constituents, biogas contains smaller concentrations of impurities, including hydrogen sulfide (H_2S), water vapour (H_2O), ammonia (NH_3), oxygen (O_2), and siloxanes.

Since methane is the component responsible for the calorific value, the relatively high concentration of carbon dioxide and other contaminants limits the direct use of raw biogas in many applications. Consequently, purification processes are required to increase its methane content and produce biomethane with properties comparable to those of natural gas (Swinbourn et al. (2024)).

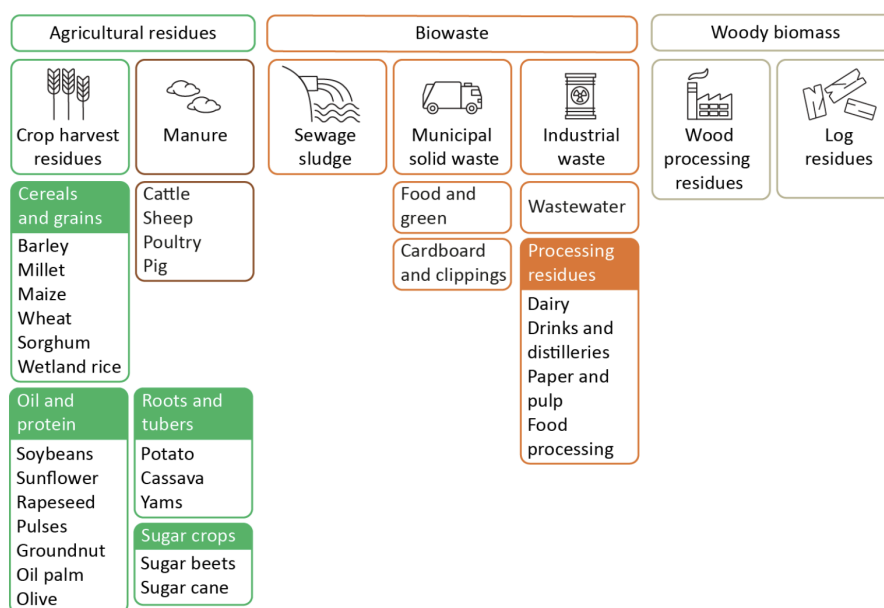


Figure 2.1: Sources of organic waste feedstock for biogas production. Source: [International Energy Agency \(2020\)](#)

2.2 Biogas Upgrading

Biogas upgrading, also referred to as biogas purification, biogas refinement or biogas methanation, is the process of increasing the methane concentration of raw biogas by removing carbon dioxide and other undesirable contaminants. The resulting methane-rich gas, known as biomethane, has a significantly higher calorific value and can be used as a substitute for natural gas in a wide range of applications. In general, biomethane should contain at least 95% methane to comply with the quality requirements for grid injection and other energy applications ([Swinbourn et al. \(2024\)](#)).

Several upgrading technologies have been developed to achieve this objective, each presenting different advantages, limitations, and operating requirements. Since a detailed comparison of these technologies is beyond the scope of this dissertation, only a brief overview of the main biogas upgrading methods is provided in the following section.

2.2.1 Biogas Upgrading Technologies

The most widely adopted methods can be broadly classified into adsorption, membrane separation, cryogenic separation, and absorption processes ([Naquash et al. \(2025\)](#)):

- **Adsorption separation:** One of the most widely used methods for biogas upgrading. Adsorption processes can be further classified into Pressure Swing Adsorption (PSA), Temperature Swing Adsorption (TSA), and Electric Swing Adsorption (ESA), with PSA being the most widely adopted. In PSA systems, biogas is compressed and passed through one or more columns filled with an adsorbent material. The adsorbent selectively retains contaminants while methane passes through the column. Once saturated, the column is regenerated

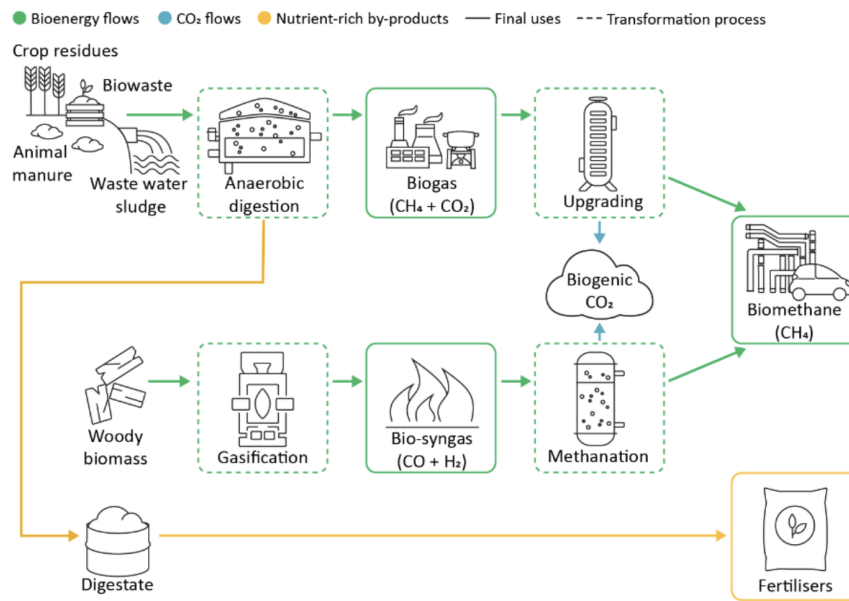


Figure 2.2: Biogas upgrading and utilization. Source: [International Energy Agency \(2020\)](#)

by depressurization, allowing the adsorbed gases to be released before starting a new adsorption cycle.

- **Membrane separation:** This method uses semi-permeable membranes, typically made of polymeric materials, to separate methane from carbon dioxide and other impurities according to their different permeation rates. The membrane material and configuration depend on the desired biomethane purity.
- **Cryogenic separation:** Cryogenic upgrading exploits the different boiling points of the gas components. By progressively reducing the temperature, carbon dioxide and other contaminants condense before methane, allowing their separation.
- **Absorption separation:** Absorption processes use a liquid solvent, such as water or chemical solvents, to selectively absorb carbon dioxide from biogas. Water scrubbing is one of the most common upgrading technologies, taking advantage of the significantly higher solubility of carbon dioxide compared to methane under standard operating conditions.

Figure 2.3 summarizes the four main biogas upgrading technologies discussed in this section.

Among the available technologies, a PSA biomethane upgrading plant was considered in this dissertation. Since the proposed AI models are developed using data collected from a PSA-based upgrading system, the following section presents a more detailed description of its operating principles and process configuration.

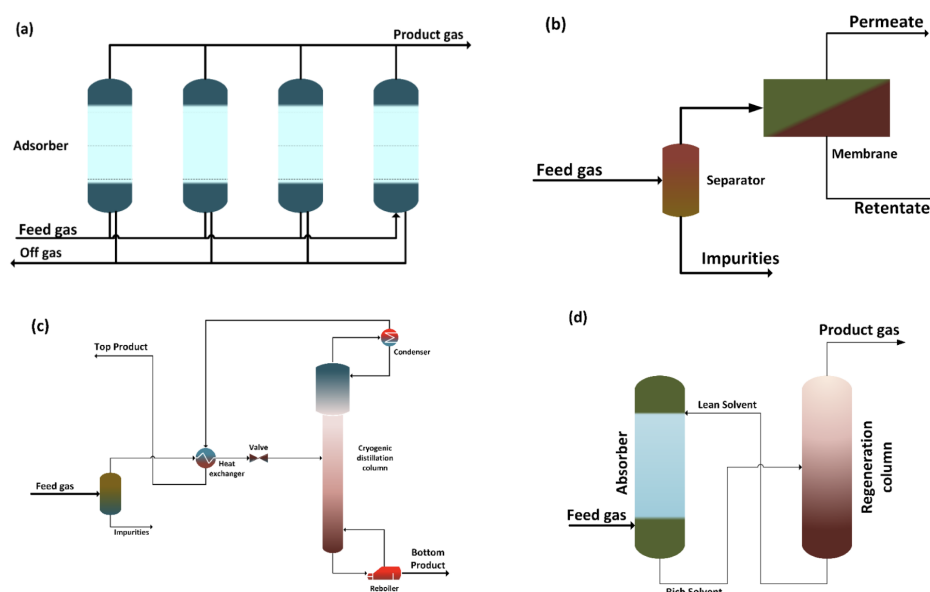


Figure 2.3: Main biogas upgrading technologies: adsorption, membrane separation, cryogenic separation, and absorption. Source: Naquash et al. (2025).

2.3 Considered Biomethane Plant

As discussed in the previous section, the biomethane upgrading plant considered in this dissertation employs PSA technology. More specifically, the plant operates using a four-column adsorption cycle and incorporates a vacuum pump during the regeneration stage to enhance the desorption of the adsorbed gases. This configuration is commonly referred to as Vacuum Pressure Swing Adsorption (VPSA).

A photograph of an example biomethane upgrading plant used in this work is presented in Figure 2.4.



Figure 2.4: Industrial VPSA biomethane upgrading plant. Adapted from Sysadvance

2.3.1 VPSA Process

VPSA separates gas components by exploiting their different affinities for porous adsorbent materials under varying pressure conditions. As previously explained, raw biogas is primarily composed of methane (CH_4) and carbon dioxide (CO_2), together with smaller amounts of water vapour and other contaminants. Before entering the adsorption unit, the biogas must first be dried to remove the water vapour. This step is essential because the adsorbent materials used inside the adsorption columns, one of which is ceramic-based, exhibit a strong affinity for water. Since the adsorption process is exothermic, excessive moisture can lead to overheating and permanent damage to the adsorbent material.

After the drying stage, the biogas is compressed before entering the adsorption columns. Operating at elevated pressures increases the probability of gas molecules interacting with the adsorbent material, improving the separation efficiency. Each adsorption column is filled with adsorbent materials specifically selected to preferentially retain carbon dioxide while allowing methane to pass through. As a result, a methane-rich stream leaves the column and is collected as biomethane.

As the adsorption process continues, the adsorbent gradually becomes saturated with carbon dioxide and loses its separation capacity. At this point, the column must be regenerated before it can be reused. Instead of stopping the production process, VPSA systems employ multiple adsorption columns operating simultaneously at different stages of the cycle. In the plant considered in this work, four adsorption columns operate in a synchronized sequence, ensuring continuous biomethane production while one or more columns undergo regeneration.

Figure 2.5 illustrates the operating principle of the four-column VPSA process. During the adsorption stage, compressed biogas enters one of the columns, where carbon dioxide is selectively adsorbed while methane exits as the product gas. Once the adsorbent approaches saturation, the column undergoes a pressure equalization stage, in which part of the compressed gas is transferred to another column operating at a lower pressure. This step recovers part of the compression energy and improves the overall efficiency of the process.

The column is then depressurized and enters the regeneration stage. Unlike conventional PSA systems, VPSA technology employs a vacuum pump to further reduce the pressure below atmospheric conditions. The low pressure promotes the desorption of the retained carbon dioxide, regenerating the adsorbent material and restoring its adsorption capacity. Since a small amount of methane is also desorbed during the initial seconds of the vacuum stage, this gas is recovered and redirected to the biogas inlet to minimize methane losses.

Finally, the regenerated column is gradually repressurized through another pressure equalization step, followed by pressure build-up until the operating pressure is reached. The column is then ready to begin a new adsorption cycle. By continuously alternating the operating stage of each column, the VPSA system is able to produce biomethane without interrupting the upgrading process.

In practice, only one adsorption column is in the production stage at any given instant, while

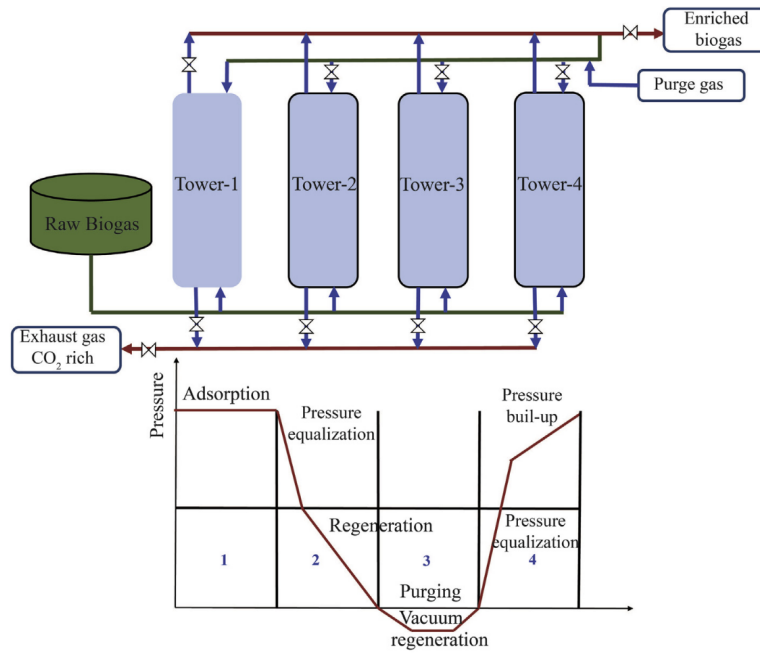


Figure 2.5: Operating cycle of a four-column VPSA system. Source: [Shah et al. \(2021\)](#)

the remaining columns are simultaneously undergoing pressure equalization, regeneration, or repressurization ([Shah et al. \(2021\)](#)).

2.3.2 Vacuum Pump

As discussed in the previous section, the vacuum pump plays a fundamental role in the VPSA process by enabling the regeneration of the adsorption columns. By reducing the column pressure below atmospheric conditions, the vacuum pump promotes the desorption of the retained gases, restoring the adsorption capacity of the adsorbent material and allowing the adsorption cycle to continue. Consequently, the performance of the vacuum pump has a direct impact on the efficiency, productivity, and methane recovery of the entire biomethane upgrading process.

Due to its critical role in the plant operation, the vacuum pump is extensively monitored in the considered biomethane plant. Several process variables, including pressure, temperature, electrical current, power consumption, and rotational speed, are continuously acquired to assess its operating condition and detect potential abnormal behaviour.

Since this dissertation focuses on the development of AI-based models for AD, FD, and PdM, the vacuum pump was selected as the primary component under analysis. Figure 2.6 shows a photograph of a working vacuum pump inside a biogas upgrading plant.

2.3.2.1 Components

The vacuum pump, whose manufacturer cannot be disclosed due to confidentiality agreements, operates according to the rotary vane principle. Oil plays a key role in the pump operation by seal-



Figure 2.6: Vacuum pump inside a biogas upgrading plant. Adapted from Sysadvantage

ing the clearances between moving parts, providing lubrication, and dissipating the heat generated during gas compression. Figure 2.7 illustrates the operating principle of the pump, showing the gas flow path during the regeneration stage of the VPSA cycle. Gas extracted from the adsorption column first enters the vacuum pump and is subsequently discharged either back to the biogas inlet or to the atmosphere, depending on the operating stage.

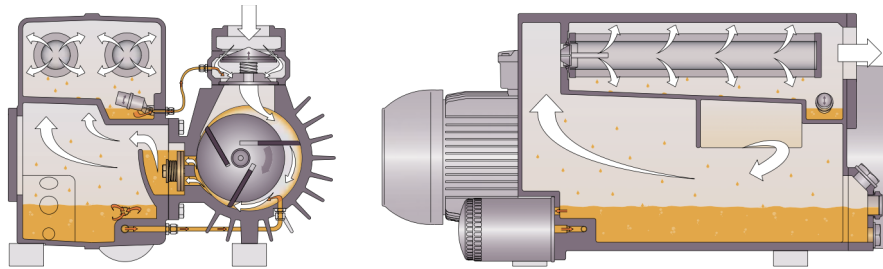


Figure 2.7: Vacuum pump operating principle. Adapted from the manufacturer's manual

A schematic representation of the pump is also presented in Figure 2.8, where the main internal components can be identified. Since a detailed mechanical description of the pump is beyond the scope of this dissertation, only the components relevant for understanding the monitored process variables are summarized in Table 2.1.

Table 2.1: Main vacuum pump components considered in this dissertation. Adapted from Sysadvantage

Abbreviation	Component	Abbreviation	Component
TSA	Resistance Thermometer	PSA	Pressure Transmitter
TS	Temperature Switch	PS	Pressure Switch
LS	Level Switch	IF	Inlet Filter
EF	Exhaust Filter	OF	Oil Filter
OS	Oil Separator	FV	Float Valve
AHE	Air-Oil Heat Exchanger	GB	Gas Ballast Valve

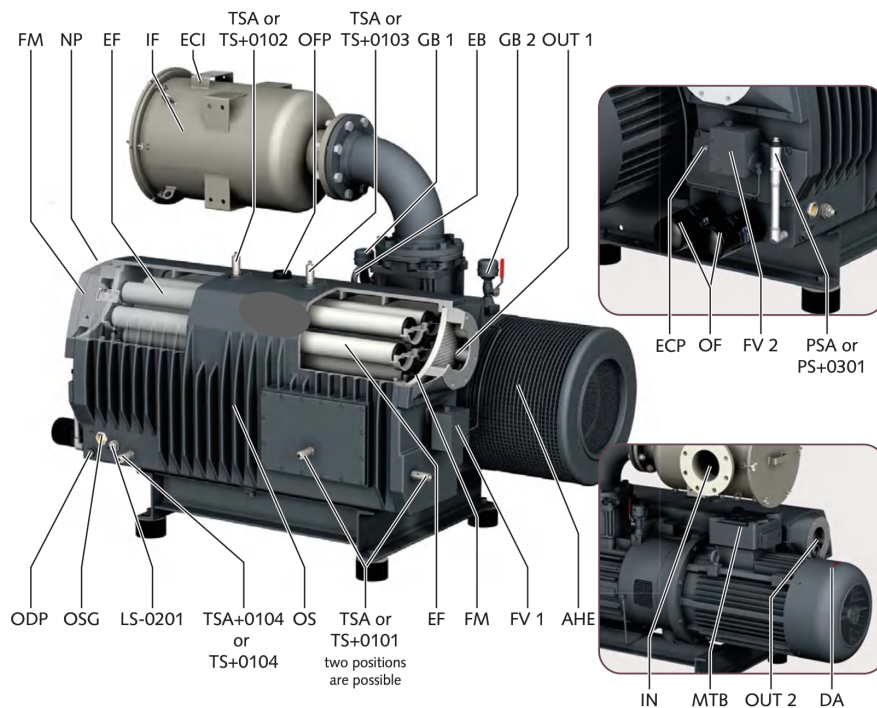


Figure 2.8: Vacuum pump scheme. Adapted from the manufacturer's manual

- **Gas Ballast Valve (GB):** Introduces a controlled amount of ambient air into the compression chamber to prevent the condensation of vapour inside the pump.
- **Inlet Filter (IF):** Protects the vacuum pump by preventing dust and other solid particles from entering the compression chamber.
- **Exhaust Filter (EF):** Removes oil aerosols from the discharged gas before it is released.
- **Oil Filter (OF):** Removes contaminants from the lubricating oil.
- **Oil Separator (OS):** Separates the lubricating oil from the discharged gas, allowing the recovered oil to be recirculated into the lubrication system.
- **Air-Oil Heat Exchanger (AHE):** Dissipates the heat generated during pump operation by cooling the lubricating oil.
- **Float Valve (FV):** Regulates the oil return from the oil separator to the compression chamber while preventing gas from flowing backwards through the lubrication circuit.
- **Resistance Thermometer (TSA):** Continuously measures the temperature of the gas or lubricating oil, providing analog temperature measurements for process monitoring.
- **Temperature Switch (TS):** Monitors the gas and oil temperatures and generates warning or trip signals when predefined temperature thresholds are exceeded.

- **Pressure Transmitter (PSA):** Continuously measures the pressure inside the oil separator, providing analog pressure measurements used for process monitoring.
- **Pressure Switch (PS):** Monitors the pressure inside the oil separator and activates warning or trip signals when predefined pressure limits are reached.
- **Level Switch (LS):** Monitors the lubricating oil level inside the oil reservoir.

2.3.2.2 Instrumentation

To ensure safe and reliable operation, the vacuum pump is equipped with several internal sensors that continuously monitor its operating conditions. These sensors measure key process variables, such as pressure, temperature, and oil level, providing real-time information about the pump's health.

In addition to continuous measurements, most sensors are associated with configurable *warning* and *trip* thresholds. A warning signal indicates that a monitored variable is approaching an abnormal operating condition. A trip signal corresponds to a critical threshold at which the pump is automatically stopped to prevent equipment damage or unsafe operation.

Table 2.2 summarizes the monitored variables together with their corresponding sensors and the associated warning and trip thresholds.

Table 2.2: Vacuum pump instrumentation and associated protection thresholds. Adapted from the manufacturer's manual.

Sensor	Measured Variable	Warning	Trip
PSA/PS 0301	Oil separator pressure	0.4 bar	0.5 bar
TSA/0101	Gas temperature	135 °C	140 °C
TSA/0102	Gas Temperature	105 °C	110 °C
TSA/0103	Gas Temperature	105 °C	110 °C
TSA/0104	Oil temperature	90 °C	110 °C

In addition to the instrumentation presented above, the vacuum pump also provides the following monitored variables, which are not associated with configurable warning or trip thresholds:

- **Oil Level Sensor:** Monitors the lubricating oil level inside the pump. Only reports whether the oil level is sufficient or low.
- **Pump Current Sensor:** Measures the electrical current drawn by the pump motor, expressed in amperes (A).
- **Pump Power Sensor:** Measures the electrical power consumed by the pump motor, expressed in watts (W).
- **Pump RPM Sensor:** Measures the rotational speed of the pump shaft, expressed in revolutions per minute (RPM).

- **Pump Speed Sensor:** Measures the operating speed of the pump as a percentage of its nominal speed.

2.3.2.3 Maintenance & Troubleshooting

Understanding the recommended maintenance procedures and the most common failure possibilities of the vacuum pump is essential for this dissertation. Besides ensuring reliable operation of the equipment, this information provides the basis for the fault scenarios considered throughout this work.

According to the manufacturer's manual, preventive maintenance should be performed at regular operating intervals to ensure the reliability and service life of the machine. Table 2.3 summarizes the recommended maintenance schedule for the main components.

Table 2.3: Recommended preventive maintenance schedule for the vacuum pump. Adapted from the manufacturer's manual.

Task	Maintenance Interval
Inspect oil level	Daily
Check machine for oil leaks	Monthly
Check the inlet filter cartridge. Change it if necessary	Monthly
Clean the machine from dust and dirt	Every 6 months
Clean the Gas Ballast Valve	Every 6 months
Check and/or clean the Air-Heat Exchanger	Every 6 months
Change the oil, oil filter and the exhaust filter	Max every 4000 hours, at the latest after 1 year

In addition to the recommended maintenance procedures, the manufacturer provides a troubleshooting guide describing the most common faults, their possible causes, and the corresponding corrective actions. This information is presented in the following Table 2.4.

Among the various faults identified by the manufacturer, a subset was selected for fault injection in a experimental dataset. These faults were chosen because they represent common operating problems that can be reproduced by modifying the monitored process variables while remaining consistent with the physical behaviour of the vacuum pump.

Table 2.4: Common troubleshooting procedures for the vacuum pump. Adapted from the manufacturer's manual.

Problem	Possible Cause	Remedy
The machine does not start.	The motor is not supplied with the correct voltage.	Check the power supply.
	The motor is defective.	Replace the motor.
	The coupling (CPL) is defective.	Replace the coupling (CPL).
The machine does not reach the specified vacuum level.	Oil level too low.	Top up the oil.
	The inlet screen (IS) is partially clogged.	Clean the inlet screen (IS).
	The inlet filter cartridge is partially clogged.	Replace the inlet filter cartridge.
The machine runs very noisily.	Worn coupling (CPL).	Replace the coupling (CPL).
	Stuck vanes.	Contact the manufacturer for repair.
	Defective bearings.	Contact the manufacturer for repair.
The machine overheats.	Insufficient cooling.	Remove dust and dirt from the machine. Check the cooling fan.
	Ambient temperature too high.	Observe the permitted ambient temperature.
	Oil level too low.	Top up the oil.
	The exhaust filters (EF) are partially clogged.	Replace the exhaust filters (EF).
The machine fumes or expels oil droplets through the gas discharge.	The exhaust filters (EF) are partially clogged.	Replace the exhaust filters (EF).
	An exhaust filter (EF) O-ring is not fitted properly.	Ensure the correct position of the exhaust filters (EF) and the O-rings.
	The float valve (FV) does not operate properly.	Check the float valve and the oil return line. Contact the manufacturer if necessary.

Chapter 3

State of the Art

This chapter reviews the theoretical foundations and the current state of the art relevant to this dissertation. It begins by introducing the fundamental concepts of artificial intelligence, machine learning, and deep learning, providing the background required to understand the methodologies adopted in this work. It then discusses the main industrial monitoring and maintenance strategies, namely anomaly detection, fault detection, and predictive maintenance. Finally, the chapter reviews the most relevant literature on deep learning for industrial time-series analysis and multi-task learning, highlighting the research gap addressed by this dissertation.

3.1 What is an AI System?

Before reviewing the state of the art related to this dissertation, it is first necessary to define what constitutes an AI system.

According to the AI Act Regulation (EU) 2024/1689 of the European Parliament ([European Data Protection Supervisor \(2025\)](#)), we can define an AI system as a system that is machine-based, can operate with some degree of autonomy, may show signs of adaptiveness after its deployment, and from a given input it receives, generates an output that can influence its environment. These outputs can be predictions, content, recommendations, or decisions.

From an academic perspective, [Russell and Norvig \(2016\)](#) define AI in terms of intelligent agents, describing them as systems that perceive their environment through sensors and act upon that environment through actuators. Under this view, an AI system can be regarded as a function that maps observations of the environment to actions intended to achieve specific goals.

Although these definitions originate from different contexts, they share the same fundamental principle: an AI system receives information from its environment, processes it, and generates outputs intended to achieve one or more predefined objectives.

3.2 Machine Learning

As discussed, AI systems are designed to perceive their environment, process information, and produce outputs depending on their objective. However, AI is a broad field that encompasses a wide range of techniques, from rule-based systems to modern data-driven approaches (Russell and Norvig (2016)).

ML is one of the most important branches of AI. Rather than relying on explicitly programmed rules, ML enables computers to automatically learn patterns and relationships from data, allowing them to improve their performance on a given task through experience. This paradigm has become the dominant approach to AI in recent decades due to the increasing availability of large datasets and advances in computational resources (Mitchell (2013); Goodfellow et al. (2016)).

According to Mitchell (2013), *"a computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."* This definition emphasizes that learning is not achieved by explicitly programming every possible decision, but rather by allowing the algorithm to discover useful patterns directly from data.

Today, ML constitutes the foundation of most modern AI systems. Depending on the complexity of the task and the amount of available data, different learning algorithms can be employed, ranging from traditional statistical methods to deep neural networks. The following sections introduce the main ML paradigms and describe the typical workflow followed during the development of ML models.

3.2.1 ML Paradigms

Machine learning algorithms can be broadly categorized according to the information available during training. The four main learning paradigms are supervised learning, unsupervised learning, reinforcement learning, and self-supervised learning. Although they share the common objective of extracting useful information from data, they differ in the learning signal available to the algorithm, the optimization objective, and the types of tasks they are designed to solve (Murphy (2022)).

3.2.1.1 Supervised Learning

Supervised learning is the most widely adopted machine learning paradigm and is used when a dataset contains both the input variables, commonly referred to as *features*, *covariates*, or *predictors*, and their corresponding desired outputs, known as *labels*. The goal is to learn a mapping between the input variables and the corresponding output so that the model can accurately predict the labels of previously unseen observations (Murphy (2022)).

Depending on the nature of the target variable, supervised learning problems are generally divided into two main categories. When the target variable belongs to a finite set of discrete classes, the problem is known as *classification*. Typical applications include fault diagnosis, image

recognition, and spam detection. Conversely, when the target variable is continuous, the problem is referred to as *regression*, where the objective is to predict numerical quantities such as energy consumption, or product demand.

Table 3.1 presents an example of a supervised learning dataset. Each observation contains four input features describing the physical characteristics of an Iris flower, together with the corresponding species label. The objective is to learn a model capable of predicting the flower species (Figure 3.1) from the measured features.

Table 3.1: Example of the Iris dataset used for supervised classification. Adapted from [Murphy \(2022\)](#)

index	Sepal Length	Sepal Width	Petal Length	Petal Width	label
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
50	7.0	3.2	4.7	1.4	Versicolor
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
149	5.9	3.0	5.1	1.8	Virginica



(a)



(b)



(c)

Figure 3.1: Three types of Iris flowers: Setosa, Versicolor and Virginica. Adapted from [Murphy \(2022\)](#)

In the context of our work, supervised learning is commonly employed for tasks such as AD, FC and PdM, where historical operational data together with fault labels or degradation measurements are available.

3.2.1.2 Unsupervised Learning

Unlike supervised learning, unsupervised learning does not deal with datasets that contain labels. Consequently, the learning algorithm must discover patterns, structures, relationships or other type of useful information directly from the data without being provided with the desired output ([Murphy \(2022\)](#)).

Common unsupervised learning tasks include *clustering*, where similar observations are grouped according to their characteristics, *dimensionality reduction*, which aims to represent high-dimensional

data using a smaller number of informative variables, and *density estimation*, which models the probability distribution of the data.

3.2.1.3 Reinforcement Learning

Reinforcement learning addresses sequential decision-making problems, where an intelligent agent learns by interacting with an environment rather than from a fixed dataset (Murphy (2022)). At each time step, the agent observes the current state of the environment, selects an action according to its policy, and receives a numerical reward indicating the quality of the chosen action. Based on this interaction, the environment transitions to a new state, and the process is repeated.

The objective of reinforcement learning is to learn an optimal policy that maximizes the expected cumulative reward over time. Since the consequences of an action may only become apparent after several subsequent decisions, the agent must balance *exploration*, by trying new actions to acquire knowledge about the environment, and *exploitation*, by selecting the actions that are currently believed to yield the highest reward.

Figure 3.2 illustrates the reinforcement learning framework. The agent continuously interacts with the environment by observing its current state, selecting an action, and receiving both a reward and the next state. Through this trial-and-error process, the agent gradually learns a policy that maximizes the expected cumulative reward.

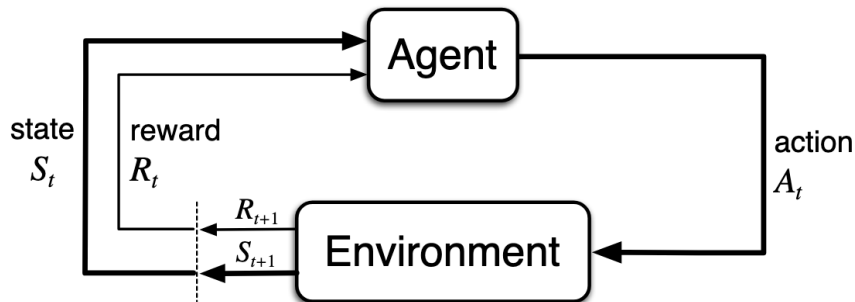


Figure 3.2: Agent–environment interaction in reinforcement learning. Adapted from Sutton and Barto (2018)

3.2.1.4 Self-Supervised Learning

Self-supervised learning is a learning paradigm in which supervision is obtained directly from the data itself rather than from manually annotated labels (Murphy (2022)). Instead of relying on external ground-truth annotations, the learning algorithm constructs auxiliary or *pretext* tasks whose targets are automatically generated from the input data. By solving these tasks, the model learns meaningful representations that capture the underlying structure of the data.

In recent years, self-supervised learning has become one of the most active research areas in machine learning, achieving remarkable success in computer vision, natural language processing and speech processing.

3.2.2 Representation Learning

The machine learning paradigms presented in the previous section describe the different ways in which models can learn from data. Regardless of the learning paradigm, the performance of a machine learning algorithm largely depends on the quality of the data representation used to describe the problem.

For relatively simple problems, carefully designed features may be sufficient to achieve good performance. For instance, if the objective is to distinguish between male and female speakers, a feature such as the average vocal tract length or the fundamental frequency may already provide enough information for a traditional classifier. However, many real-world problems involve highly complex and high-dimensional data, making it extremely difficult to manually define informative features. Image classification is a representative example of such complexity. A model cannot recognize an animal directly from the raw pixel values of an image. Instead, it must first learn simple representations, such as edges and textures, which are then combined to form higher-level features such as shapes, body parts, and eventually complete objects. This hierarchical learning process allows increasingly abstract representations to be constructed from simpler ones. This concept is called representation learning (Goodfellow et al. (2016)). Figure 3.3 illustrates the concept of hierarchical representation learning.

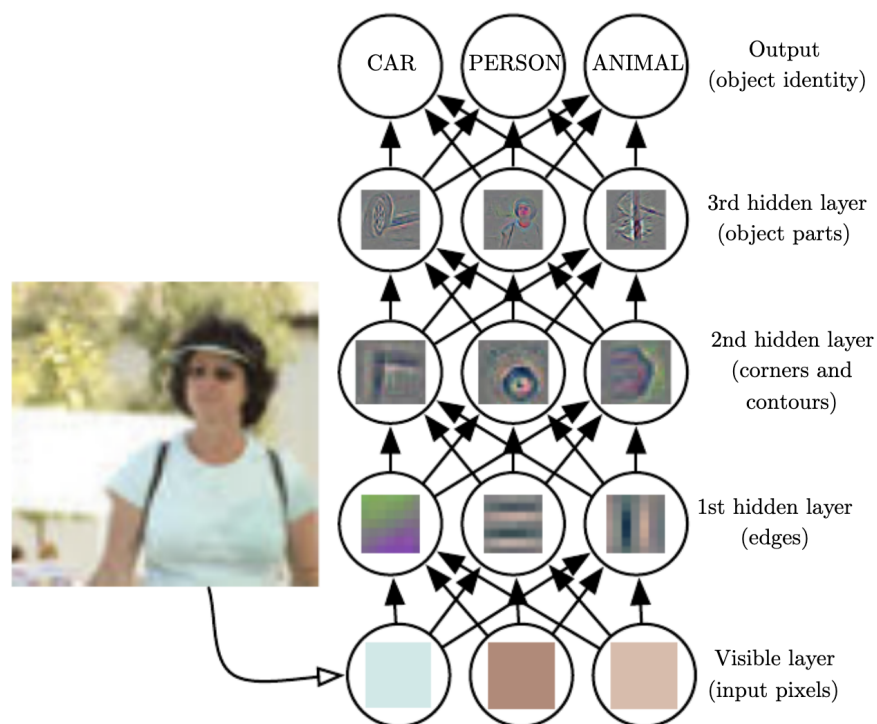


Figure 3.3: Illustration of a representation learning model. Adapted from Goodfellow et al. (2016)

3.2.3 Deep Learning

Representation learning provides the foundation for DL. Rather than relying on manually engineered features, DL algorithms automatically discover multiple levels of representation by composing a sequence of simple transformations. This hierarchical organization enables models to learn increasingly abstract features directly from raw data, making them particularly effective for solving complex tasks such as computer vision, speech recognition, and natural language processing (Goodfellow et al. (2016)).

Most modern DL models are implemented using Artificial Neural Networks (ANNs). These models are constructed by composing multiple layers of simple nonlinear transformations, allowing increasingly abstract representations of the input data to be learned automatically (Goodfellow et al. (2016)).

The following section introduces the fundamental concepts behind ANNs, which constitute the basis of most modern DL models.

3.2.3.1 Artificial Neural Networks

From a mathematical perspective, an ANN can be regarded as a parameterized function that maps an input vector $\mathbf{x}$ to an output $\mathbf{h}$. During training, the parameters of this function are optimized so that the network approximates the underlying relationship between the input data and the desired output (Goodfellow et al. (2016)).

The fundamental computational unit of an ANN is the artificial neuron, also known as a perceptron (Rosenblatt (1958)). A perceptron receives one or more input variables, computes a weighted sum of these inputs, adds a bias term, and applies a nonlinear activation function to produce its output. Mathematically, this operation can be expressed as

$$h = g(\mathbf{w}^T \mathbf{x} + b) \quad (3.1)$$

where $\mathbf{x}$ denotes the input vector, $\mathbf{w}$ is the vector of trainable weights, $\mathbf{b}$ is the bias term, and g represents the activation function. Figure 3.4 illustrates the structure of a perceptron.

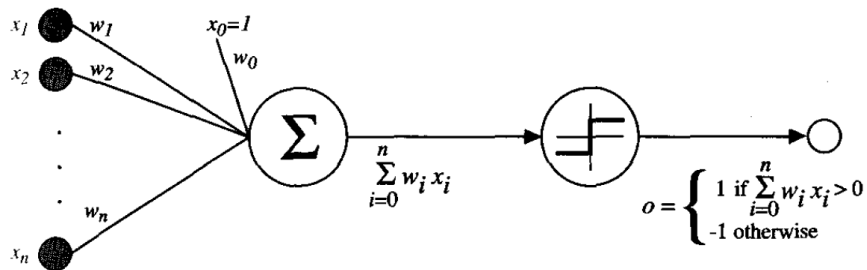


Figure 3.4: Illustration of a perceptron. Adapted from Mitchell (2013)

Although the perceptron constitutes the fundamental computational unit of an ANN, practical problems generally require much more expressive models. To achieve this, multiple perceptrons

are interconnected and organized into successive layers, forming a feedforward neural network, commonly referred to as a Multilayer Perceptron (MLP). The network typically consists of an input layer, one or more hidden layers, and an output layer. Each neuron receives the outputs of the previous layer, performs the operation defined in Equation 3.1, and forwards its output to the next layer.

By stacking multiple hidden layers, an MLP is capable of learning increasingly abstract representations of the input data, allowing it to approximate highly complex nonlinear functions (Goodfellow et al. (2016)). Figure 3.5 illustrates the general architecture of a Multilayer Perceptron.

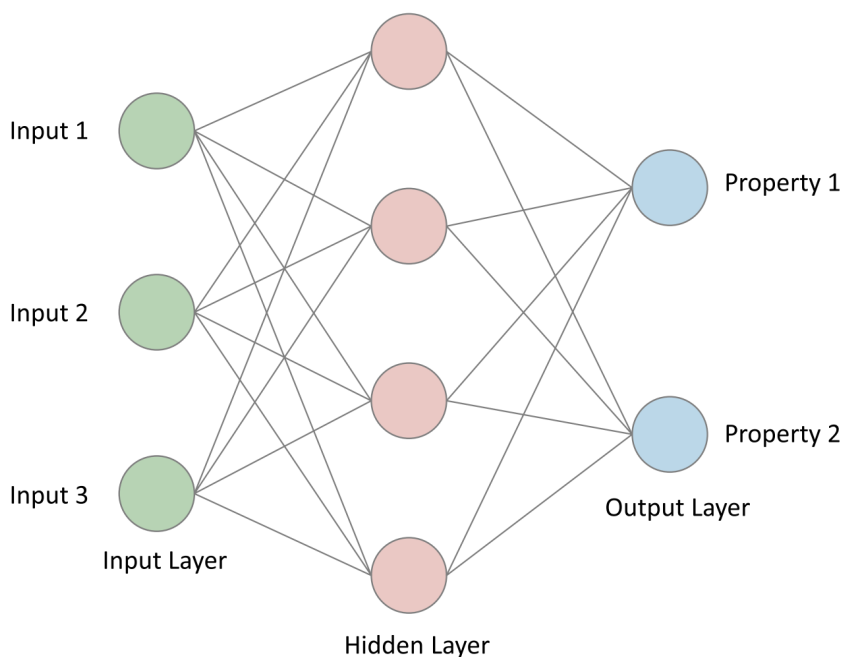


Figure 3.5: Illustration of a MLP with 1 hidden layer. Adapted from Zhang et al. (2023)

3.2.4 Training Deep Neural Networks

Once the architecture of a neural network has been defined, the next step is to determine the values of its parameters, namely the weights and biases. This process, known as training, consists of finding the set of parameters that minimizes a *cost function* $J(\theta)$, defined as the average loss incurred by the model over the training set (Goodfellow et al. (2016)). During training, the network repeatedly processes examples from the training dataset, evaluates the quality of its predictions through a *loss function*, and updates its parameters in order to reduce this cost.

In practice, the cost function is rarely evaluated over the entire training set at every update. Instead, most deep learning models are trained using *stochastic gradient descent* (SGD) or one of its variants, in which the gradient is estimated from a small, randomly sampled *mini-batch* of examples at each iteration. This trades off the accuracy of the gradient estimate for a substantial

reduction in computational cost per update, and is essential for training on large datasets (Goodfellow et al. (2016)).

This optimization process can be divided into four main stages: forward propagation, loss (cost) computation, backpropagation, and parameter update. These four stages are repeated iteratively, over mini-batches and across multiple passes (epochs) over the training set, until the model converges or a predefined stopping criterion is reached (Goodfellow et al. (2016)).

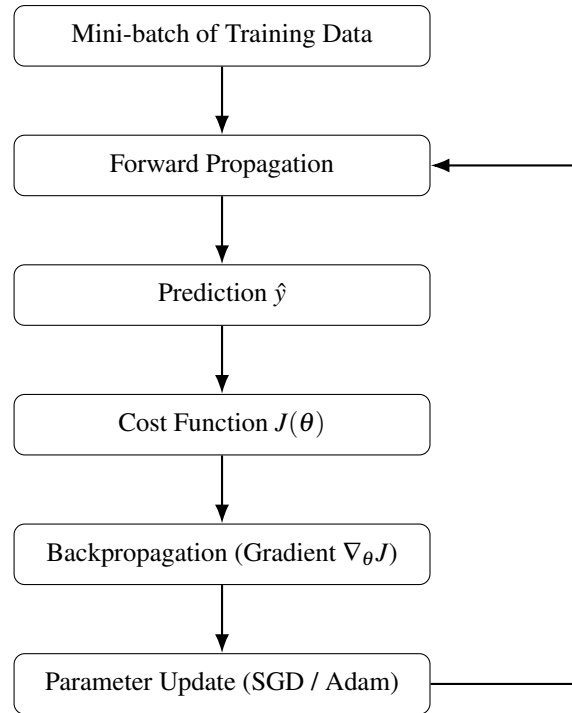


Figure 3.6: General training pipeline of a deep neural network. Backpropagation computes the gradient of the cost function; the optimization rule (e.g., SGD or Adam) then uses this gradient to update the parameters.

3.2.5 Deep Learning Architectures

Over the years, numerous deep learning architectures have been proposed to address different types of learning problems. Each architecture is designed to exploit specific characteristics of the input data and is therefore suited to different application domains.

Since this dissertation focuses on industrial time-series data, only the architectures that are directly relevant to this work are presented. In particular, this section introduces recurrent neural networks (RNNs) and their Long Short-Term Memory (LSTM) variant, which constitute the foundation of the models developed in the following chapters.

3.2.5.1 Recurrent Neural Network

Feedforward neural networks process each input independently, without retaining information from previous inputs. Consequently, they are unable to capture temporal dependencies that nat-

usually arise in sequential data. Recurrent Neural Networks (RNNs) overcome this limitation by introducing recurrent connections that allow information from previous time steps to be propagated through the network.

Unlike feedforward networks, an RNN maintains an internal hidden state that acts as a memory of past observations. At each time step, the hidden state is updated using both the current input and the hidden state from the previous time step. This enables the network to exploit temporal dependencies when making predictions (Goodfellow et al. (2016)).

$$\mathbf{h}_t = g(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}) \quad (3.2)$$

Figure 3.7 illustrates the recurrent structure of an RNN unfolded through time. The same set of parameters is shared across all time steps, allowing the network to process sequences of arbitrary length while maintaining information from previous observations.

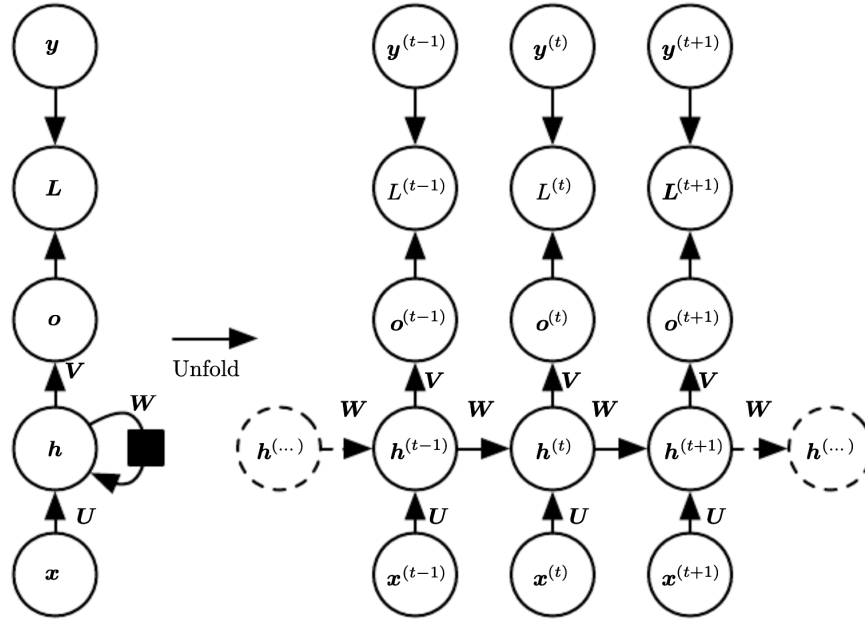


Figure 3.7: Illustration of a RNN. Adapted from Goodfellow et al. (2016)

Although RNNs are theoretically capable of modeling long-term dependencies, in practice they suffer from the vanishing and exploding gradient problems during training. As a consequence, learning information over long sequences becomes difficult. To address this limitation, the Long Short-Term Memory (LSTM) architecture was proposed (Hochreiter and Schmidhuber (1997)).

3.2.5.2 Long Short-Term Memory

Long Short-Term Memory (LSTM) networks were proposed by Hochreiter and Schmidhuber in 1997 to overcome the limitations of conventional RNNs when modeling long-term dependencies (Hochreiter and Schmidhuber (1997)). Unlike a standard RNN, an LSTM introduces a memory

cell that is specifically designed to preserve relevant information over long sequences while discarding irrelevant information.

The key component of an LSTM is its gating mechanism, which regulates the flow of information through the network. Three gates are responsible for this process: the forget gate determines which information from the previous memory should be discarded, the input gate decides which new information should be stored in the memory cell, and the output gate controls which information is propagated to the hidden state and the subsequent time step. By explicitly controlling the information flow, LSTMs mitigate the vanishing gradient problem and are therefore capable of learning much longer temporal dependencies than conventional RNNs (Goodfellow et al. (2016)).

Figure 3.8 illustrates the internal structure of an LSTM cell. At each time step, the forget, input, and output gates regulate the information stored in the cell state, allowing the network to selectively retain or discard information according to its relevance for future predictions.

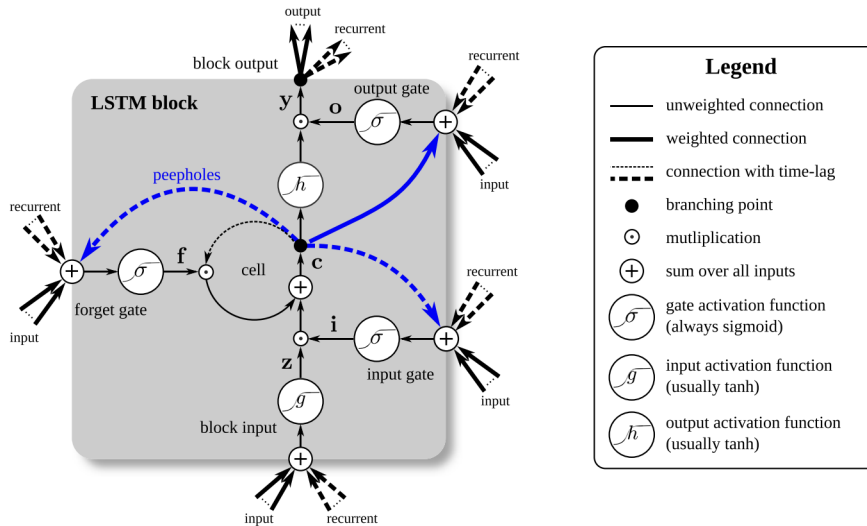


Figure 3.8: Illustration of a LSTM. Adapted from Greff et al. (2017)

3.2.6 Learning Strategies

The architectures presented in the previous sections describe how neural networks process information. However, the same architecture can be trained following different learning strategies depending on the number of tasks the model is expected to perform. The two most common approaches are single-task learning and multi-task learning (Zhang and Yang (2022)).

In single-task learning (STL), a model is trained to perform only one prediction task by optimizing a single objective function. As a result, a separate model must be developed and trained for each task of interest. In contrast, multi-task learning (MTL) trains a single model to perform multiple related tasks simultaneously. This is typically achieved by sharing part of the network among all tasks while using task-specific output layers. By learning a shared representation, MTL can exploit the common information between related tasks, often leading to improved generalization,

better data efficiency, and reduced overfitting when compared with training independent models (Zhang and Yang (2022)).

Figure 3.9 illustrates the conceptual difference between the two learning strategies. While STL requires an independent model for each prediction task, MTL relies on a shared feature extraction stage followed by multiple task-specific output branches. Figure 3.10 shows an example multi-task feedforward neural network.

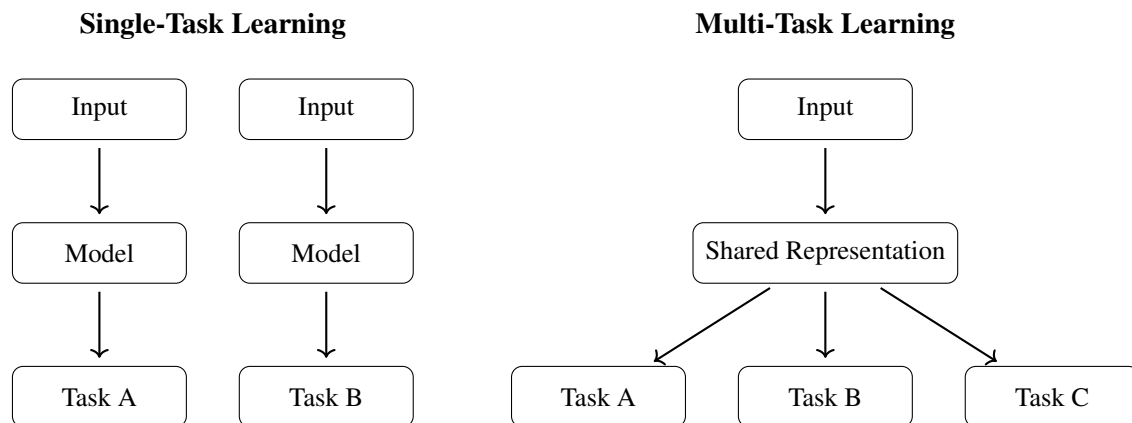


Figure 3.9: Conceptual comparison between single-task learning and multi-task learning. In STL, each task requires an independent model, whereas MTL shares a common representation across multiple related tasks while maintaining task-specific output layers.

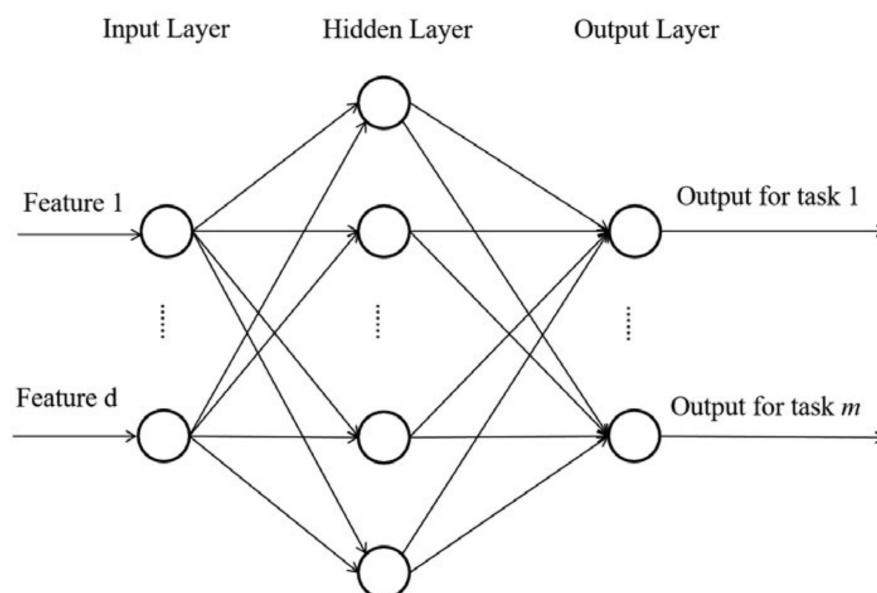


Figure 3.10: Example for the multi-task feedforward neural network with an input layer, a hidden layer, and an output layer.. Adapted from Zhang and Yang (2022)

3.3 Industrial Monitoring and Maintenance

Modern industrial systems are increasingly equipped with a large number of sensors that continuously monitor the operating conditions of equipment and production processes. These sensors generate large volumes of operational data, including measurements of pressure, temperature, flow rate, vibration, electrical variables, and other process parameters. By analyzing these data, it is possible to assess the health of industrial assets, identify abnormal operating conditions, detect faults, and anticipate potential failures before they lead to unplanned downtime.

To support these objectives, several data-driven monitoring and maintenance strategies have been developed. Among the most widely adopted are anomaly detection (AD), fault detection (FD), and predictive maintenance (PdM). Although these concepts are closely related, they address different stages of the equipment health monitoring process and provide complementary information for supporting maintenance decisions. The following sections introduce each of these concepts and discuss their relationship.

3.3.1 Anomaly Detection

Anomaly Detection (AD) is the process of identifying observations, events, or patterns that deviate significantly from the expected behaviour of a system (Chandola et al. (2009)). Such observations, commonly referred to as *anomalies* or *outliers*, are rare and differ from the majority of the data, often indicating abnormal operating conditions, equipment degradation, sensor malfunctions, or unexpected process disturbances.

In industrial environments, anomaly detection plays a fundamental role in condition monitoring by continuously analyzing sensor data to identify deviations from normal operation. Since anomalies often represent the earliest indication of an emerging problem, their timely detection enables operators to take corrective actions before faults develop into equipment failures, thereby improving system reliability and reducing unplanned downtime (Darban et al. (2025)).

Unlike fault detection, anomaly detection does not necessarily identify the cause or type of the abnormal behaviour. Instead, its primary objective is to determine whether the observed system behaviour is consistent with normal operation or represents a significant deviation that requires further investigation.

Depending on the availability of labelled data, AD can be formulated either as a supervised binary classification problem, distinguishing between *normal* and *anomalous* observations, or as an unsupervised task, in which the model learns the distribution of normal operating conditions and flags significant deviations from it without relying on labelled anomalies (Chandola et al. (2009); Darban et al. (2025)). The choice between these formulations largely depends on the availability and reliability of historical fault records.

3.3.2 Fault Detection

Fault Detection (FD) is the process of determining whether a fault has occurred in a monitored system by identifying deviations from its expected operation (Isermann (2006)). In contrast to anomaly detection, which only indicates that abnormal behaviour is present, fault detection aims to identify the existence of a specific fault and, in many cases, determine its type or location. This information enables maintenance personnel to understand the origin of the problem and select appropriate corrective actions.

Since FD is concerned with identifying which type of fault is present rather than merely flagging abnormal behaviour, it is typically formulated as a supervised multiclass classification problem, in which each observation is assigned to one of several predefined fault categories, or to a *no-fault* class representing normal operation (Isermann (2006)). This formulation requires access to historical data containing examples of the specific faults of interest, which are often obtained either from real fault occurrences or, as in the case of this dissertation, through controlled fault injection.

3.3.3 Predictive Maintenance

Predictive Maintenance (PdM) is a maintenance strategy that uses historical and real-time operational data to assess the health condition of industrial equipment and predict future failures before they occur (Carvalho et al. (2019)). Unlike traditional maintenance strategies, such as corrective maintenance, which is performed after a failure occurs, or preventive maintenance, which follows predefined maintenance schedules, PdM seeks to estimate the optimal time for maintenance interventions based on the actual condition of the equipment.

PdM can be formulated in different ways depending on the nature of the target variable. When the objective is to estimate a continuous quantity, such as the Remaining Useful Life (RUL) of a component or a continuous health index reflecting its degradation state, PdM is treated as a regression problem. Alternatively, when the objective is to anticipate whether a failure will occur within a predefined future time window, PdM can instead be formulated as a supervised classification problem (Carvalho et al. (2019); Khan et al. (2026)). Both formulations rely on the ability to model the temporal evolution of the equipment condition from historical sensor data.

3.3.4 Relationship Between AD, FD and PdM

Although anomaly detection, fault detection, and predictive maintenance are closely related, they address different stages of the equipment health monitoring process. Anomaly detection focuses on identifying abnormal behaviour without necessarily determining its cause. Once an anomaly has been detected, fault detection aims to identify the existence and, when possible, the type of fault responsible for the abnormal behaviour. Predictive maintenance extends this process by estimating the future evolution of the equipment condition, enabling maintenance actions to be scheduled before failures occur.

Despite addressing different aspects of equipment health monitoring, AD, FD, and PdM share a common underlying structure: all three tasks can be framed as supervised learning problems, whether classification or regression, applied to the same multivariate time-series data collected from process sensors. This shared structure is what motivates the multi-task learning approach adopted in this dissertation, in which a single model exploits a shared representation of the sensor data to jointly address the three tasks, rather than training a separate model for each one (Section 3.2.6).

3.4 Related Work

3.4.1 Deep Learning for Industrial Monitoring

Anomaly Detection. AD in industrial settings is typically formulated either as a supervised binary classification problem or, more commonly, as an unsupervised task in which a model learns the distribution of normal operating data and flags significant deviations from it. According to a systematic review of the field, convolutional, recurrent, autoencoder-based, and hybrid architectures have proven particularly effective for this task, although persistent challenges remain, including limited labelled fault data and class imbalance (Zaidi et al. (2026)). Reconstruction-based approaches are especially prevalent: a model is trained exclusively on normal operating data, and an observation is classified as anomalous once its reconstruction error exceeds a predefined threshold (Khan et al. (2026)).

Fault Detection. FD extends AD by requiring the identification of the specific fault type, and is therefore generally formulated as a supervised multiclass classification problem. Convolutional architectures are among the most widely adopted for this stage, frequently combined with recurrent networks to jointly capture spatial and temporal patterns in the sensor signals (Khan et al. (2026); Zaidi et al. (2026)).

Predictive Maintenance. PdM is commonly addressed either as a regression problem, estimating the Remaining Useful Life (RUL) of a component, or as a classification problem, predicting the likelihood of failure within a future time window. Recurrent architectures, particularly LSTM networks, remain the dominant choice for this stage, often combined with convolutional layers for feature extraction prior to temporal modelling (Khan et al. (2026)). However, most publicly available datasets used to develop and evaluate these models are generated in controlled laboratory conditions, limiting their representativeness of real-world industrial degradation processes (Khan et al. (2026)).

3.4.2 Multi-Task Learning for Industrial Monitoring

Beyond single-task approaches, several works have proposed multi-task architectures that jointly address two of the three monitoring stages discussed above. Yin (2026) developed a multi-task framework for semiconductor manufacturing systems, employing an LSTM-based autoencoder as

a shared backbone to jointly perform AD and RUL prediction, explicitly addressing the trade-off between the two tasks during training. Similarly, Yan et al. (2023) proposed a multi-task architecture combining a shared feature extraction module with a Multi-gate Mixture-of-Experts (MMoE) layer to jointly perform health-state assessment and RUL prediction on sensor-equipped machines.

Both works follow a common architectural pattern, consisting of a shared representation module coupled with task-specific output branches, consistent with the multi-task learning strategy introduced in Section 3.2.6. However, neither work combines all three monitoring stages, AD, FD, and PdM, within a single model, and neither is applied to a biomethane upgrading context.

3.4.3 Research Gap

The literature reviewed in the previous sections reveals that the vast majority of studies addressing industrial condition monitoring focus on a single task in isolation, whether anomaly detection, fault detection, or remaining useful life estimation, each typically supported by a dedicated deep learning model (Khan et al. (2026); Zaidi et al. (2026)). Although deep learning has achieved remarkable performance on these individual problems, comparatively few works have investigated multi-task learning strategies capable of jointly addressing multiple monitoring objectives within a single model. The multi-task approaches that do exist are generally limited to combining two of these tasks, such as anomaly detection and remaining useful life estimation (Yin (2026)), or health-state assessment and remaining useful life estimation (Yan et al. (2023)), rather than jointly addressing anomaly detection, fault detection, and predictive maintenance within a unified framework.

Furthermore, the application of these deep learning and multi-task learning approaches to biomethane upgrading plants remains, to the best of our knowledge, entirely unexplored in the literature, despite the growing industrial relevance of this sector. In particular, no prior work has addressed AI-based condition monitoring for the vacuum pump, a critical asset in the VPSA process considered in this dissertation.

These observations motivate the development and evaluation of a multi-task deep learning framework capable of jointly performing anomaly detection, fault detection, and predictive maintenance for the vacuum pump of a biomethane upgrading plant, and of comparing its performance against both single-task deep learning models and traditional machine learning baselines.

Chapter 4

Conclusions and Future Work

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4.1 Results

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4.2 Further Work

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4.3 Impact on Sustainable Development Goals

Add in this section a reflection on how your work impacts the United Nations's Sustainable Development Goals (SDG). If you want to read about the SDG, visit the page: <https://sdgs.un.org/goals>. To obtain more information on the subject, we suggest the following courses:

- <https://dl4sd.org/course/view.php?id=4>
- <https://www.classcentral.com/course/global-sustainable-development-10311>

Appendix A

Loren Ipsum

If you are going to use a passage of Lorem Ipsum, you need to be sure there isn't anything embarrassing hidden in the middle of text.

A.1 What is *Loren Ipsum*?

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum¹.

A.2 Where does Loren come from?

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, *consectetur*, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of “*de Finibus Bonorum et Malorum*” (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, “*Lorem ipsum dolor sit amet...*”, comes from a line in section 1.10.32.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from “*de Finibus Bonorum et Malorum*” by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

¹ Available at <http://www.lipsum.com/>

A.3 Why using Loren?

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using “Content here, content here”, making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for “lorem ipsum” will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

A.4 Where to Find Examples?

There are many variations of passages of Lorem Ipsum available, but the majority have suffered alteration in some form, by injected humour, or randomised words which don't look even slightly believable. If you are going to use a passage of Lorem Ipsum, you need to be sure there isn't anything embarrassing hidden in the middle of text. All the Lorem Ipsum generators on the Internet tend to repeat predefined chunks as necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with a handful of model sentence structures, to generate Lorem Ipsum which looks reasonable. The generated Lorem Ipsum is therefore always free from repetition, injected humour, or non-characteristic words etc.

Appendix B

Transparency on the Use of AI

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